

Bacterial Non-Homologous End Joining: The Ku/LigD Module

A review-style synthesis of the compact prokaryotic system for Ku/LigD-mediated repair of DNA double-strand breaks

1. Executive Summary

Bacterial non-homologous end joining (NHEJ) is the minimal, self-contained pathway by which many prokaryotes reseal DNA double-strand breaks (DSBs) without a homologous template. In stark contrast to the multi-protein eukaryotic apparatus (Ku70/Ku80, DNA-PKcs, Artemis, XRCC4/Ligase IV, XLF, PAXX, and others), the bacterial system is built from just **two dedicated proteins**: a homodimeric **Ku** DNA-end-binding factor and a multifunctional DNA ligase, **LigD**. Together these two players carry out every activity the reaction requires — end recognition, end protection, end bridging (synapsis), end processing (fill-in, resection, and nucleotide "healing"), and ATP-dependent sealing ([PMID: 17506672](#); [PMID: 12215643](#)).

The best-supported mechanistic model is straightforward and sequential: Ku homodimers thread onto and slide along broken duplex ends, protect them from nucleolytic degradation, bridge the two ends into a synaptic complex, and recruit LigD via a species-variable Ku C-terminus. LigD then uses three autonomous catalytic domains — an archaeo-eukaryotic-primase (AEP)-like **polymerase (POL)**, a manganese-dependent **3'-phosphoesterase (PE)**, and an **ATP-dependent ligase (LIG)** — to remodel incompatible ends into a ligatable configuration and seal the backbone ([PMID: 16023671](#); [PMID: 16446439](#)). The pathway is template-free and intrinsically **error-prone**, leaving molecular scars — single-nucleotide (frequently ribonucleotide) insertions, templated fill-ins, and short deletions — at the junctions ([PMID: 15778718](#)).

Three broad conclusions emerge from this investigation. First, bacterial NHEJ is **physiologically specialized for the non-replicating state** — stationary phase, dormancy, and spores — precisely the conditions where a sister chromatid is unavailable and homologous recombination (HR) cannot operate ([PMID: 17360246](#); [PMID: 17496093](#)). Second, the module is assembled from **ancient, redeployed protein folds** — a Ku core shared with eukaryotes and a

polymerase from the AEP primase clan — and is distributed patchily across bacteria through gene loss and horizontal transfer, even being pirated by mycobacteriophages ([PMID: 11483577](#); [PMID: 16949369](#)). Third, the "minimal" two-component system is flanked by **backup and alternative routes** — LigC/LigC1 ligases, the PrimC primase, and a faithful, Ku/LigD-independent joining of cohesive 3'-overhangs — so that end structure and physiological context, not a single fixed enzyme set, dictate outcome ([PMID: 18281464](#); [PMID: 24957619](#)).

2. Definition and Biological Boundaries

2.1 What is included

The system reviewed here is the **species-neutral Ku/LigD module** for DSB repair by NHEJ in bacteria. Its defining features are:

- **Ku** — a prokaryotic, homodimeric DNA-end-binding protein that recognizes broken double-stranded DNA ends in a sequence-independent manner, protects and aligns them, and recruits the ligase.
- **LigD** — a dedicated, modular NHEJ enzyme that provides end-processing (polymerase and nuclease/phosphoesterase) and ATP-dependent ligation activities within a single polypeptide.
- The **functional coupling** between them: Ku-bound, synapsed ends provide the substrate context for LigD-mediated processing and sealing ([PMID: 16023671](#); [PMID: 20018881](#)).

2.2 What is excluded and what is commonly confused

Several neighboring processes are frequently discussed alongside bacterial NHEJ but are mechanistically distinct and lie outside this scope:

Process	Why it is separate
Homologous recombination (HR) — RecA/RecBCD/AdnAB	Template-dependent, error-free, requires a homologous donor; dominant in replicating cells. HR and NHEJ are parallel, partly redundant DSB routes; loss of one is compensated by the other (PMID: 21219454; PMID: 24658131).
Single-strand annealing (SSA)	A third, RecA-independent DSB route in mycobacteria that uses RecBCD; distinct machinery and logic (PMID: 21219454).
Mismatch repair, nucleotide excision repair, base excision repair	Repair single-strand lesions, not DSBs. Note that some NHEJ proteins <i>moonlight</i> in BER (Ku AP-lyase activity; LigD/LigC/PrimC in BER), which can blur assays but does not change the DSB-repair definition (PMID: 25355514; PMID: 33918798).
Eukaryotic NHEJ	Shares a Ku core and functional logic but requires many additional factors (DNA-PKcs, XRCC4-LigIV, XLF, Artemis). The bacterial system is a two-protein reduction, not a scaled-down copy (PMID: 17506672).

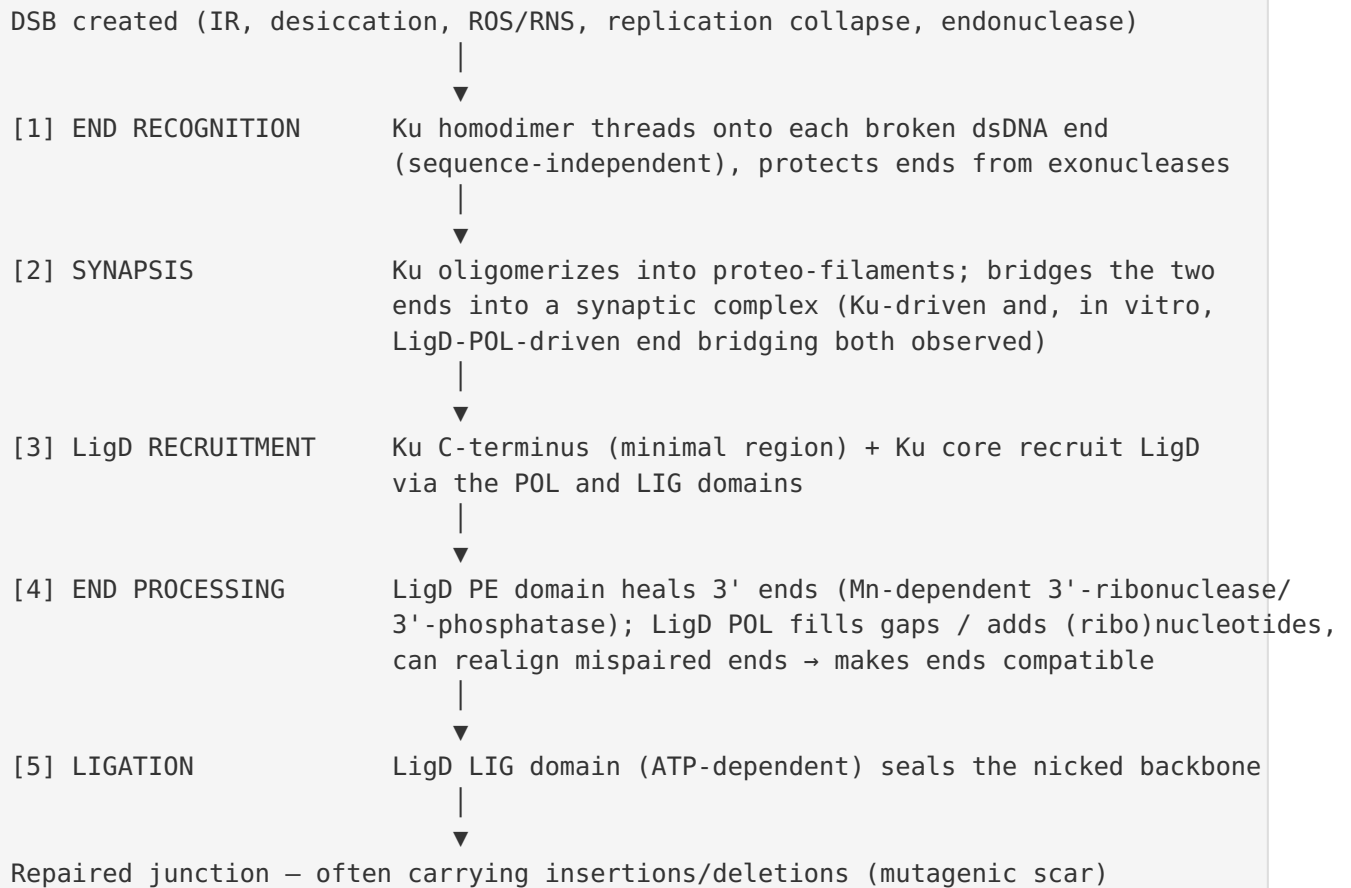
2.3 Competing definitions

The literature is largely consistent in defining bacterial NHEJ as the Ku + LigD two-component system (PMID: 17938628). The main definitional ambiguity concerns **how expansive "the pathway" is**. A strict definition limits it to Ku + LigD. A broader, more accurate definition recognizes that in organisms such as *Mycobacterium smegmatis*, NHEJ is a **network** in which the LigD POL domain can partner with backup ligases (LigC1) or stand-alone POL-domain paralogs (PolD1/PolD2), and in which accessory factors (a Sir2-like deacetylase) modulate efficiency (PMID: 24957619; PMID: 23198659; PMID: 21637345). This review adopts the two-protein core as the boundary while explicitly cataloguing the accessory and backup layer.

3. Mechanistic Overview

3.1 The canonical sequence of events

The best current model runs as follows:



3.2 Obligatory, conditional, and accessory steps

- **Obligatory:** DNA-end recognition/protection by Ku and phosphodiester sealing by an ATP-dependent ligase (normally LigD LIG) are required for repair of blunt and 5'-overhang breaks in vivo; deleting *ku* or *ligD* abolishes NHEJ and sensitizes non-replicating cells to ionizing radiation (PMID: 21219454; PMID: 12215643).
- **Conditional:** End processing (POL fill-in, PE healing) is invoked only when ends are not directly ligatable. Blunt and 5'-overhang breaks depend stringently on the POL domain; cohesive 3'-overhangs can bypass Ku/LigD entirely (PMID: 18281464).
- **Accessory:** Backup ligases (LigC1), stand-alone POL paralogs (PolD1/PolD2), and modulators (Sir2-like deacetylase) contribute redundancy and regulation but are individually dispensable under standard conditions (PMID: 24957619; PMID: 23198659; PMID: 21637345).

4. Major Molecular Players and Active Assemblies

4.1 Key Finding 1 — Bacterial NHEJ is a minimal, autonomous two-component system (Ku + LigD)

Bacterial NHEJ collapses the eukaryotic multi-factor apparatus into just two proteins that between them possess all the break-recognition, end-processing, and ligation activities required. This was first predicted bioinformatically by Aravind & Koonin (2001), who noted that prokaryotic Ku homologs form predicted operons with genes for an ATP-dependent DNA ligase and/or an archaeo-eukaryotic-type primase — the signature of a self-contained DSB-repair cassette ([PMID: 11483577](#)). The authoritative microbial review of Pitcher, Brissett & Doherty (2007) states the point directly: *"the bacterial NHEJ complex is a two-component system that, despite its relative simplicity, possesses all of the break-recognition, end-processing, and ligation activities required to facilitate the complex task of DSB repair"* ([PMID: 17506672](#)). Genetics confirm the functional minimalism: ablating *ku* or *ligD* abolishes NHEJ and renders non-replicating mycobacteria hypersensitive to ionizing radiation ([PMID: 21219454](#); [PMID: 24957619](#)).

4.2 Key Finding 2 — LigD is a modular enzyme with three autonomous catalytic domains

LigD is the "Swiss-army knife" of the system. Domain dissection of *M. tuberculosis* LigD demonstrated that its polymerase (primase), nuclease/phosphoesterase, and ligase functions reside in physically separable, individually active domains ([PMID: 16023671](#)). Weller et al.'s and later structural work established the modular architecture directly: *"LigD consists of an ATP-dependent ligase domain fused to a polymerase domain (Pol) and a phosphoesterase module"* ([PMID: 16446439](#)). The three modules are:

Domain	Fold / family	Activity	Notes
POL (polymerase)	AEP / archaeal-primase-like	Templated & non-templated (ribo)nucleotide addition; gap fill; end bridging	Two-metal mechanism, Mn-dependent, prefers rNTPs; the direct catalyst of mutagenic NHEJ in vivo
PE (3'-phosphoesterase)	Novel bifunctional phosphoesterase family	Mn-dependent 3'-ribonuclease + 3'-phosphatase ("end healing")	Resects 3'-ribonucleotide tracts; requires penultimate 2'-OH for the ribonuclease step
LIG (ligase)	ATP-dependent DNA ligase	Nick sealing	Requires active-site lysine for adenylation

The crystal structure of the *Pseudomonas/Mycobacterium* LigD POL domain revealed the archaeal-primase-like fold and two-metal chemistry, and — importantly — showed that "*a mutation of Mycobacterium LigD that uniquely ablates the polymerase activity results in increased fidelity of blunt-end double-strand break repair in vivo by virtue of eliminating nucleotide insertions at the recombination junctions*," pinning the source of NHEJ mutagenesis on the POL domain (PMID: 16446439). The PE domain's Mn-dependent 3'-ribonuclease/3'-phosphatase resection chemistry was mapped in a series of *Pseudomonas* studies (PMID: 15897197; PMID: 16540477; PMID: 16046407).

4.3 Key Finding 3 — Bacterial Ku shares a conserved core with eukaryotic Ku but has a unique, multipurpose C-terminus

Aravind & Koonin dissected Ku into a **eukaryote/prokaryote-conserved core ring** plus lineage-specific extensions — "*the Ku core that is conserved in eukaryotes and prokaryotes*" (PMID: 11483577). Bacterial Ku lacks the eukaryotic von Willebrand A (vWA) domain and functions as a **homodimer** (versus the Ku70/Ku80 heterodimer of eukaryotes). The bacterial-specific **C-terminus** is the functional hotspot: dissection in *Bacillus subtilis* showed that "*the minimal C-terminus is required for the Ku-LigD interaction, whereas the basic extension controls the threading and DNA bridging abilities of Ku*" (PMID: 26961308). In *M. tuberculosis*, the Ku C-terminus is a "multipurpose arm" that binds DNA and LigD and stimulates ligation, with the extended C-terminus limiting DNA binding and the Ku core also contributing to ligase stimulation and direct LigD binding (PMID: 36250639). Recent cryo-EM work resolved apo and

DNA-bound mycobacterial Ku homodimers and showed that **Ku forms extended proteo-filaments upon binding DNA**, driving DNA synapsis; the C-terminus regulates DNA loading and subsequent LigD recruitment (PMID: 41298423; PMID: 41521670).

4.4 Key Finding 7 — The Ku-recruits-ligase core mechanism is strongly supported experimentally

The founding experimental demonstration is Weller et al. (2002, *Science*), which established that bacterial Ku homologs retain the end-binding biochemistry of eukaryotic Ku, that *"bacterial Ku specifically recruits DNA ligase to DNA ends and stimulates DNA ligation. Loss of these proteins leads to hypersensitivity to ionizing radiation in Bacillus subtilis"* (PMID: 12215643). This Ku-recruits-and-stimulates-ligase mechanism has since been reproduced across *Mycobacterium*, *Pseudomonas*, and *Bacillus* by genetics, in-vitro reconstitution, and structural biology (PMID: 15778718; PMID: 20018881; PMID: 23691176; PMID: 36250639), and is synthesized in the definitive Shuman & Glickman review, which confirms that *"mycobacteria and other genera have an NHEJ system that depends on a dedicated DNA ligase, LigD, and the DNA-end-binding protein Ku"* (PMID: 17938628).

5. Evolutionary and Cell-Biological Variation

5.1 Key Finding 4 — Dedicated to non-replicating/dormant states, and intrinsically mutagenic

Bacterial NHEJ is not a general-purpose DSB pathway; it is the **repair option of last resort in cells that cannot use HR**. In *M. smegmatis*, deletion of *ku/ligD* sensitizes cells to ionizing radiation and desiccation **specifically in stationary phase, not in log phase** — because in replicating cells a sister chromatid supports error-free HR: *"NHEJ-mutant strains (Deltaku, DeltaligD, Deltaku/ligD) were significantly more sensitive to ionizing radiation (IR) during stationary phase than wild-type M. smegmatis. However, there was little difference in IR sensitivity between NHEJ-mutant and wild-type strains in logarithmic phase"* (PMID: 17360246; PMID: 17496093). NHEJ is the **sole** DSB pathway operating during *M. tuberculosis* dormancy (PMID: 41521670).

The price of template-free joining is **fidelity**. Repair of blunt and 5'-overhang DSBs is roughly 50% mutagenic, with molecular signatures of templated fill-in, non-templated single-nucleotide additions (often ribonucleotides), and nucleolytic resection: *"NHEJ of blunt-end and complementary 5'-overhang DSBs is highly mutagenic"* (PMID: 15778718). At chromosomal I-SceI breaks, NHEJ produces predominantly single-nucleotide insertions and rapidly mutates the break site in the absence of HR (PMID: 17496093).

5.2 Key Finding 5 — End structure dictates pathway choice; backup ligases and a faithful alternative route exist

The system is best understood as **end-structure-tuned**. Aniukwu et al. (2008) showed that repair of blunt and 5'-overhang breaks depends stringently on Ku and the LigD POL domain, whereas 3'-overhang breaks are handled by *"a novel end-joining pathway specific to breaks with 3' overhangs that is Ku- and LigD-independent and perfectly faithful"* — most plausibly relying on the essential NAD⁺-dependent replicative ligase LigA (PMID: 18281464). Redundancy extends to ligases: LigC1 supports a **LigD-ligase-independent backup route that still requires the LigD POL domain** — *"we define a LigD ligase-independent NHEJ pathway in Mycobacterium smegmatis that requires the ATP-dependent DNA ligase LigC1 and the POL domain of LigD"* (PMID: 24957619); LigC also provides backup for blunt-end repair (PMID: 15778718); and LigC plus the AEP-primase PrimC contribute additional backup activities under oxidative stress (PMID: 33918798).

Pathway-choice summary:

End structure	Primary route	Ku required?	LigD required?	Fidelity
Blunt	Ku + LigD (POL-dependent)	Yes	Yes (POL critical)	Mutagenic
5'-overhang	Ku + LigD (POL fill-in)	Yes	Yes	Mutagenic
Cohesive 3'-overhang	LigA-dependent, Ku/LigD-independent	No	No	Faithful
Blunt (backup)	LigC / LigC1 + LigD POL	Yes	POL only	Variable

5.3 Key Finding 6 — Ancient, redeployed folds; patchy distribution via loss and horizontal transfer

The Ku/LigD module is a mosaic of old parts. The **prokaryotic Ku core is homologous to the eukaryotic Ku core**, placing the fold before the eukaryote/prokaryote divergence; eukaryotic Ku later acquired a vWA domain and heterodimerized (PMID: 11483577). The **LigD polymerase belongs to the archaeo-eukaryotic primase (AEP) superfamily**, adopting "*a fold similar to that of archaeal DNA primase*" (PMID: 16446439; PMID: 17174332). The two genes frequently sit together in predicted operons — "*genes for prokaryotic Ku homologs form predicted operons with genes coding for an ATP-dependent DNA ligase and/or archeal-eukaryotic-type DNA primase*" (PMID: 11483577). The module is **absent from many bacteria** (e.g., *E. coli*), giving a patchy phyletic distribution consistent with gene loss and horizontal transfer (PMID: 17506672). Strikingly, mycobacteriophages Corndog and Omega carry their **own Ku** and hijack the host LigD to circularize their genomes via short 4-base cos ends, demonstrating the module's mobility and modularity (PMID: 16949369).

Because the LigD polymerase is the representative bacterial member of the AEP clan and Ku's core is the representative shared fold, these two domains — rather than the lineage-specific C-terminal extensions or accessory paralogs — are the best proxies for understanding the ancestral NHEJ role.

6. Constraints, Dependencies, and Failure Modes

Ordering constraints. The reaction is strongly ordered: Ku must load and protect ends *before* recruiting LigD (Ku is upstream and rate-limiting), and end **processing precedes ligation** — the PE domain heals 3' termini and the POL domain fills/realigns before the LIG domain can seal (PMID: 16023671; PMID: 17947582). In *M. tuberculosis*, Ku is the rate-limiting factor for the whole pathway (PMID: 41521670).

Physiological / compartment constraints. NHEJ is functionally confined to the **non-replicating state**. In log phase, HR outcompetes and largely replaces it; the two pathways are partly redundant so that inactivating NHEJ is compensated by elevated HR, and vice versa (PMID: 21219454; PMID: 24658131). This physiological gating is why NHEJ mutants show phenotypes only in stationary phase, spores, and dormancy (PMID: 17360246; PMID: 31948638).

Substrate constraints. End chemistry governs which route is used and whether processing is needed. Faithful, direct joining is possible for compatible cohesive ends; blunt/5'-overhang ends require mutagenic POL activity; the PE 3'-ribonuclease step even requires a specific 2'-OH configuration on the penultimate ribose (PMID: 18281464; PMID: 16540477).

Evidence ruling out otherwise-plausible paths. The Ku/LigD-independent, perfectly faithful 3'-overhang route (PMID: 18281464) rules out the assumption that *all* bacterial end joining flows through Ku/LigD. The POL-ablation experiment (PMID: 16446439) rules out the LIG or PE domains as the principal source of insertional mutagenesis. Stand-alone POL paralogs PolD1/PolD2, although biochemically LigD-POL-like, were shown *not* to substitute for LigD POL in fill-in during NHEJ under the conditions tested, ruling them out as the primary NHEJ polymerases (PMID: 23198659).

Failure modes. Loss of Ku or LigD → radiation/desiccation hypersensitivity of dormant cells; loss of both HR and NHEJ ($\Delta ku \Delta ligD \Delta recA$) cripples intracellular survival of *M. tuberculosis*, an effect rescuable by neutralizing host ROS/RNS (PMID: 24658131). Because NHEJ is mutagenic, its "success" can itself be a failure mode at the sequence level, fixing insertions/deletions at the repaired locus (PMID: 17496093).

7. Controversies and Open Questions

- 1. What drives synapsis in vivo — Ku filaments or the LigD polymerase?** Recent cryo-EM shows Ku itself oligomerizes into proteo-filaments that bridge ends (PMID: 41298423; PMID: 41521670), whereas an earlier crystal structure captured the **LigD POL domain mediating end synapsis** as a proposed bridging intermediate (PMID: 17947582). Whether these are sequential stages, redundant mechanisms, or context-dependent alternatives is unresolved.
- 2. What is the source of repair-associated deletions?** The POL domain explains insertions, and the PE/nuclease activity can generate deletions in reconstituted *E. coli* systems (PMID: 20530153), but the in-vivo origin of terminal deletions at junctions is not fully assigned.
- 3. How general is the mycobacterial/Pseudomonas paradigm?** Nearly all mechanistic detail comes from *Mycobacterium*, *Pseudomonas*, and *Bacillus*. Given the module's patchy distribution and species-variable Ku C-termini and LigD domain arrangements (some organisms split LigD activities across separate proteins), extrapolating a single model across all NHEJ-competent bacteria risks overgeneralization (PMID: 17506672).

4. **Are second Ku–LigD contact sites functionally important?** Beyond the primary POL–Ku interaction, evidence for a second Ku-dependent LIG-domain interaction has been reported (PMID: 20530153; PMID: 36250639); their relative in-vivo weight is unclear.
5. **Role of accessory modulators.** A Sir2-like NAD⁺-dependent deacetylase interacts with Ku and LigD and modestly promotes NHEJ (PMID: 21637345); the regulatory logic (e.g., acetylation control of end binding) is not established.

Well-supported vs. indirect. Strongly supported: the two-component composition, LigD's three-domain modular chemistry, Ku's end-binding/recruitment role, POL as the mutagenesis catalyst, and the stationary-phase specialization — all backed by convergent genetics, biochemistry, and structures. More indirect / model-dependent: the exact in-vivo synaptic architecture, the LigA identity of the faithful 3'-overhang route (inferred), and the physiological importance of backup ligases outside stress conditions.

8. Limitations and Knowledge Gaps

- **Organism bias.** The evidence base is dominated by a handful of Actinobacteria and *Bacillus/Pseudomonas*; broad phyla with Ku/LigD homologs remain experimentally unexplored.
- **In-vitro vs. in-vivo gap.** Much end-processing chemistry (PE ribonuclease specificity, POL ribonucleotide preference) is characterized on defined primer-templates that may not reflect chromosomal DSB contexts.
- **Reconstitution artifacts.** Heterologous reconstitution in *E. coli* (which lacks native NHEJ) is powerful but can over- or under-represent activities (e.g., nuclease-driven deletions) relative to the native host.
- **Structural snapshots.** The new cryo-EM Ku-filament and older POL-synapsis structures are static; the temporal choreography of loading → synapsis → processing → ligation is inferred, not directly observed as a trajectory.

9. Proposed Follow-up Experiments / Actions

1. **Time-resolved / single-molecule synapsis assays** (e.g., smFRET or live-cell imaging as in PMID: 41521670) comparing Ku-only, LigD-only, and Ku+LigD conditions to determine

whether Ku filaments and LigD-POL bridging are sequential or alternative synaptic mechanisms.

2. **Cross-phylum functional survey.** Reconstitute Ku/LigD pairs from underexplored phyla in a common heterologous host to test how far the mycobacterial paradigm generalizes and to catalogue split-LigD architectures.
 3. **Deletion-source mapping.** Use domain-specific catalytic mutants (PE-dead, POL-dead, nuclease-dead) at a defined chromosomal I-SceI break to assign the enzymatic origin of junction deletions in vivo.
 4. **Genetic test of the faithful 3'-overhang route.** Directly test LigA (and other candidate ligases) dependence for Ku/LigD-independent cohesive-end joining to confirm the inferred LigA route ([PMID: 18281464](#)).
 5. **Regulatory dissection of accessory factors.** Define whether the Sir2-like deacetylase regulates Ku/LigD by acetylation, and whether backup ligases (LigC/LigC1) and PrimC are induced under specific stresses ([PMID: 21637345](#); [PMID: 33918798](#)).
 6. **Therapeutic targeting.** Given NHEJ's essentiality in *M. tuberculosis* dormancy, pursue Ku-synapsis or LigD-domain inhibitors as adjuncts that block persistence-phase DSB repair ([PMID: 41298423](#)).
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10. Key References

PMID	Contribution
12215643	Weller et al. 2002, <i>Science</i> — founding identification of the bacterial NHEJ complex; Ku recruits and stimulates ligase; IR sensitivity in <i>B. subtilis</i> .
11483577	Aravind & Koonin 2001 — bioinformatic prediction of the prokaryotic DSB-repair system; conserved Ku core; Ku–ligase/AEP-primase operons.
17506672	Pitcher, Brissett & Doherty 2007 — microbial-perspective review defining the two-component system.
17938628	Shuman & Glickman 2007 — authoritative review of bacterial NHEJ (Ku + LigD).
16023671	Pitcher et al. 2005 — LigD domain dissection (POL, PE, LIG).
16446439	Zhu et al. 2006 — POL crystal structure; AEP-primase fold; POL as in-vivo mutagenesis catalyst.
17174332	Brissett et al. 2007 — Mt-LigD PolDom structure; AEP superfamily; 5'-phosphate recognition.
15897197 , 16540477 , 16046407	Zhu & Shuman — PE domain 3'-ribonuclease/3'-phosphatase chemistry.
26961308	McGovern et al. 2016 — <i>B. subtilis</i> Ku C-terminus: LigD recruitment vs bridging/threading.
36250639	Sowa et al. 2022 — Mt-Ku C-terminus as multipurpose arm.
41298423 , 41521670	Cryo-EM of mycobacterial Ku; proteo-filament synapsis model.
15778718	Gong et al. 2005 — low-fidelity NHEJ; LigC backup.
17360246 , 17496093	Stationary-phase specificity; mutagenic chromosomal repair.
18281464	Aniukwu et al. 2008 — end-structure-dependent pathway choice; faithful Ku/LigD-independent 3'-overhang route.
24957619	Bhattarai et al. 2014 — LigC1 backup NHEJ requiring LigD POL.
21219454	Gupta et al. 2011 — three genetically distinct DSB pathways (HR, NHEJ, SSA).

PMID	Contribution
16949369	Mycobacteriophage-encoded Ku; module mobility.
17947582	POL-mediated DNA synaptic complex structure.
20018881	<i>Pseudomonas</i> LigD gap-filling; Ku stimulation and end protection.
23691176	<i>B. subtilis</i> minimal NHEJ reconstituted in vitro.
20530153	Mt-Ku/Mt-LigD reconstitution in <i>E. coli</i> ; nuclease-driven deletions; second Ku–LIG contact.
23198659	PolD1/PolD2 stand-alone POL paralogs.
21637345	Sir2-like deacetylase in mycobacterial NHEJ.
33918798	LigC/PrimC backup under oxidative stress.
24658131	HR or NHEJ sufficient for intracellular <i>M. tuberculosis</i> survival.
25355514	Ku AP-lyase (BER moonlighting) activity.
31948638	Ku/LigD in <i>B. subtilis</i> spore resistance to electron-beam treatment.

Prepared as a commissioned review synthesis of the Ku/LigD bacterial NHEJ module. Mechanistic detail is drawn predominantly from Mycobacterium, Pseudomonas, and Bacillus; claims should not be assumed to generalize uniformly to all NHEJ-competent bacteria.